

BERNARDO SUBERCAEAUX

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EDUCATION

Carnegie Mellon University, PA, United States August 2021 - (expected) 2027
Ph.D. in Computer Science, advised by Marijn J. H. Heule.

Thesis: *SAT Encodings: From Art to Science*.

Committee: Ryan Williams (MIT), Stefan Szeider (TU Wien), Jeremy Avigad (CMU), and Ruben Martins (CMU).

University of Chile, Chile March 2014 - December 2020

Master in Computer Science (GPA 7.0/7.0), under the supervision of Pablo Barceló and Jorge Pérez.

Double Bachelor in Engineering and Computer Science (GPA 7.0/7.0), 2019.

1st place in Latin American Contest of Master's theses in Artificial Intelligence IEEE LA-CCI.

CentraleSupélec (ex École Centrale Paris), France September 2016 - June 2018

Bachelor + Master in Engineering - Computer Science (GPA 3.58/4.00)

Eiffel Excellence Scholarship (sole Chilean recipient, 2016).

RESEARCH SUMMARY

My research spans automated reasoning, algorithms, and computational complexity. I use SAT solvers to tackle long standing open problems in discrete mathematics via finite computation, especially via the design of novel *SAT encodings*; that is, reformulations of the problem that yield speed-ups of several orders of magnitude. Lately, I've been developing a theory of the possibilities and limits of SAT encodings in terms of their size and propagation strength, with a particular focus on how efficient encodings can be found automatically

CONFERENCE PUBLICATIONS

30. **LPAR 2026**, *A SAT Attack on Tarski's High School Algebra Problem*, with B. Przybocki.
29. **LPAR 2026**, *Coloring Queens with Thousands of Encodings*, with B. Przybocki and M. Heule.
28. **CICM 2026**, *Doubly Saturated Ramsey Graphs: A Case Study in Computer-Assisted Mathematical Discovery*, with B. Przybocki, J. Mackey, and M. Heule.
27. **SAT 2026**, *Automated Reencoding Meets Graph Theory*, with B. Przybocki and M. Heule.
26. **SAT 2026**, *Near-Optimal Encodings of Cardinality Constraints*, with A. Krapivin and B. Przybocki.
25. **STOC 2026**, *Optimal and Efficient Partite Decompositions of Hypergraphs*, with A. Krapivin, B. Przybocki, and N. Sanhueza-Matamala.
24. **FUN 2026**, *A Demigod's Number for the Rubik's Cube*, with A. Merino.
23. **FUN 2026**, *Solving Small Rubik's Cubes as Slowly as Possible*, with J. Quan, N. Kim, and J. Mackey.
22. **FUN 2026**, *Price of Locality in Permutation Mastermind: Are TikTok Influencers Chaotic Enough?* (single author).
21. **PODS 2025** (**Distinguished paper award, invited to special issue of TODS**), *Explaining k -Nearest Neighbors: Abductive and Counterfactual Explanations*, with P. Barceló, A. Kozachinskiy, M. Romero, and J. Verschae.
20. **CICM 2025** (**Best paper award**), *Automated Symmetric Constructions in Discrete Geometry*, with E. Mackey, L. Qian, and M. Heule.
19. **CADE 2025**, *Unfolding Boxes with Local Constraints*, with L. Qian, E. Wang, and M. Heule.
18. **INFOCOM 2025**, *Latency Guarantees for Caching with Delayed Hits*, with K. Gurushankar and N. Singer.
17. **AAAI 2025**, *Probabilistic Explanations for Linear Models*, with M. Arenas and K. Meel.

16. **CG 2024**, *A Mathematical Analysis of PlaceIt: A Game of Perfect Online Sorting*, with P. Ruiz Cuevas and C. Chock.
15. **KR 2024**, *A Uniform Language to Explain Decision Trees*, with M. Arenas, P. Barceló, D. Bustamante, and J. Caraball.
14. **CICM 2024 (Best paper award runner-up)**, *Automated Mathematical Discovery and Verification: Minimizing Pentagons in the Plane*, with J. Mackey, M. Heule, and R. Martins.
13. **ITP 2024**, *Formal Verification of the Empty Hexagon Number*, with W. Nawrocki, J. Gallicchio, C. Codel, M. Carneiro, and M. Heule.
12. **LPAR 2024**, *Sometimes Hoarding is Harder than Cleaning: NP-hardness of Maximum Blocked-Clause Addition* (single author).
11. **FUN 2024**, *PackIt! Gamified Rectangle Packing*, with T. Garrison and M. Heule.
10. **LPAR 2023 (Best paper award)**, *Toward Optimal Radio Colorings of Hypercubes via SAT-solving*, with M. Heule.
9. **TACAS 2023 (Featured in Quanta Magazine, Best paper award nomination, invited to special issue of JAR)**, *The Packing Chromatic Number of the Infinite Square Grid is 15*, with M. Heule.
8. **SAT 2022**, *The Packing Chromatic Number of the Infinite Square Grid is at Least 14*, with M. Heule.
7. **NeurIPS 2022**, *Augmenting Online Algorithms with ε -Accurate Predictions*, with A. Gupta, D. Panigrahi, and K. Sun.
6. **NeurIPS 2022**, *On Computing Probabilistic Explanations for Decision Trees*, with M. Arenas, P. Barceló, and M. Romero.
5. **FUN 2022**, *Wordle is NP-hard*, with D. Lokshtanov.
4. **NeurIPS 2021 (Spotlight, < 3%)**, *Foundations of Symbolic Languages for Model Interpretability*, with M. Arenas, D. Baez, P. Barceló, and J. Pérez.
3. **FUN 2021**, *The Computational Complexity of Evil Hangman*, with J. Barbay.
2. **NeurIPS 2020**, *Model Interpretability through the Lens of Computational Complexity*, with P. Barceló, M. Monet, and J. Pérez.
1. **ICDT 2020**, *On the Expressiveness of LARA: A Unified Language for Linear and Relational Algebra*, with P. Barceló, N. Higuera, and J. Pérez.

JOURNAL PUBLICATIONS

5. **TALG 2026**, *Computing NP-hard Repetitiveness Measures via MAX-SAT*, with H. Bannai, K. Goto, M. Ishihata, S. Kanda, D. Köppl, and T. Nishimoto.
4. **JAIR 2025** (journal version), *On Computing Probabilistic Explanations for Decision Trees*, with M. Arenas, P. Barceló, A. Kozachinskiy, and M. Romero.
3. **NAW 2024**, *A proof long enough to stump Leonhard Euler*, with M. Heule.
2. **TCS 2022** (journal version), *On the expressiveness of Lara: A proposal for unifying linear and relational algebra*, with P. Barceló, N. Higuera, and J. Pérez.
1. **IOI Journal 2016**, *Wavelet Trees for Competitive Programming*, with R. Castro, N. Lehmann, and J. Pérez.

OTHER PUBLICATIONS (WORKSHOPS AND PREPRINTS)

5. **arXiv preprint 2025**, *From the Finite to the Infinite: Sharper Asymptotic Bounds on Norin's Conjecture via SAT*, with M. Kirchweger, T. Peitl, and S. Szeider.
4. **ModRef 2025**, *Asymptotically Smaller Encodings for Graph Problems and Scheduling* (single author).
3. **arXiv preprint 2024**, *Pentagon Minimization without Computation*, with J. Mackey.
2. **AFCI@NeurIPS 2020**, *Foundations of Languages for Interpretability and Bias Detection*, with P. Barceló and J. Pérez.
1. **DEEM 2019 & AMW 2019**, *Expressiveness of Matrix and Tensor Query Languages in terms of ML Operators*, with P. Barceló, N. Higuera, and J. Pérez.

INVITED TALKS

- **LIACS, Leiden University** (online) August 2026
Breaking down graphs into structured pieces, optimally and efficiently.
- **MAA MathFest: Computers and Combinatorics**, Boston August 2026
SAT Solving for the Discrete Mathematician's Toolbox.
- **Dagstuhl Seminar 26221**, Dagstuhl May 2026
Compact CNF Encodings for Boolean Functions.
- **Université Clermont-Auvergne**, Clermont-Ferrand May 2026
Breaking down graphs and hypergraphs into structured pieces, optimally and efficiently.
- **Banff International Research Station**, Banff January 2026
Towards a Science of CNF Encodings.
- **AGCO Seminar**, Pontificia Universidad Católica de Chile, Santiago November 2025
Optimal d -Clique Decompositions for Hypergraphs.
- **Simons Institute / SLMATH**, Berkeley April 2025
Computer-Assisted Intuition: SAT Solvers in Mathematical Discovery.
- **Joint Mathematics Meetings: 35th Anniversary of AI and Math**, Seattle January 2025
Automated Reasoning for the Discrete Mathematician.
- **Boston Computation Club**, Boston August 2023
The Packing Chromatic Number of the Infinite Square Grid.
- **Stanford University**, Stanford October 2022
Power to computers, Power to the people: from computer-aided mathematics to formal explainability.
- **Logic-based Methods in Machine Learning (LMML)**, FLoC, Haifa July 2022
The Exciting Theory of Formal Explanations.
- **AGCO Seminar**, CMM, Universidad de Chile, Santiago July 2022
Augmenting Online Algorithms with ε -Accurate Predictions.
- **DataLab UC: Manejo de Datos y Teoría de la Computación** (online) April 2021
Árboles y diagramas de decisión: operaciones y equivalencias.
- **LINKS Seminar**, Inria / Université de Lille (online) February 2021
Foundations of Languages for Interpretability.

RESEARCH INTERNSHIPS AND VISITS

- **TU Wien**, Research visit to Prof. Stefan Szeider (Summer 2025)
- **Amazon**, Automated Reasoning Group, supervised by Mike Whalen (Summer 2024)
- **Amazon**, Automated Reasoning Group, supervised by Mike Whalen (Summer 2023)
- **Simons Institute**, Data-Driven Decision Processes (Fall 2022)

TEACHING EXPERIENCE

Teaching assistant for 18 course offerings across 11 distinct courses (CMU, CentraleSupélec, University of Chile, PUC Chile): *Logic and Mechanized Reasoning, Undergraduate Quantum Computation, Complexity and Computability, Design and Analysis of Algorithms, Discrete Math, Theory of Computation, Introduction to Programming, Algorithms and Data Structures, Databases, Programming Languages, and Workshop for Competitive Programming.*

SERVICE

Program Committee for FUN 2026 and AAAI 2025. Helped organize the ModRef workshop 2026. Reviewed for TALG, ESA, ITCS, JAIR, SAT, NeurIPS, Communications of the ACM, TACAS, ICLR, and Operations Research. In total, I have reviewed 37 papers.